

Sodium & Salt Reduction in CKD

Your most powerful daily choice for protecting your kidneys

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< 2,000 mg
sodium / day
CKD TARGET

4,500-6,000 mg
sodium / day
FILIPINO AVERAGE

1 g salt
= 400 mg sodium
THE KEY CONVERSION

Of all the things a CKD patient can change about their diet, sodium reduction has the strongest evidence and the largest measurable benefit — and it's the area where most Filipinos have the most room to improve.

Why it works so powerfully

Every 2.3 g reduction in daily sodium lowers systolic BP by 5 mmHg and reduces proteinuria by ~25% in patients on ACE inhibitors or ARBs — amplifying the effect of every kidney-protective drug you take. Each extra 1 g/day above target raises cardiovascular event risk by 17%, independent of blood pressure (PURE study, NEJM 2014).

Daily sodium targets by patient group

Patient group	Daily sodium target	Notes / salt equivalent
General CKD Stage 1-4	< 2,000 mg/day	< 5 g salt (1 tsp)
CKD with hypertension or proteinuria	< 1,500 mg/day	< 3.75 g (¾ tsp)
CKD with heart failure / edema	< 1,500 mg/day	< 3.75 g (¾ tsp)
Haemodialysis (HD)	< 2,000 mg/day	+ fluid ≤ 1 L/day
Peritoneal dialysis (PD)	2,000-2,500 mg	Still far below typical PH intake
Per-meal practical target	< 600 mg / meal	3 × 600 + snacks ≤ 200 = ~2,000 mg

Source: KDIGO 2024 CKD guideline · AHA recommendations · WHO sodium intake guideline 2012

Where the sodium actually hides in Filipino food



All values are per typical Filipino serving. The % column shows what fraction of the 2,000 mg CKD daily target each single serving consumes.

Food / item	Serving	Sodium (mg)	% of 2,000 mg
CONDIMENTS AND SAUCES			
Patis (fish sauce)	1 tbsp (15 mL)	1,420	71%
Bagoong alamang	1 tbsp	1,370	69%
Bagoong isda (sardine paste)	2 tbsp	1,500	75%
Toyo (regular soy sauce)	1 tbsp	900	45%
Toyomansi (toyo + calamansi)	1 tbsp	820	41%
Reduced-sodium toyo	1 tbsp	540	27%
Banana ketchup	1 tbsp	165	8%
Vinegar (suka)	1 tbsp	2	< 1%
Calamansi juice	1 tbsp	1	< 1%
BOUILLON, SEASONING POWDERS, INSTANT MIXES			
Knorr bouillon cube	1 cube (10 g)	1,500	75%
Maggi Magic Sarap	1 sachet (8 g)	1,200	60%
Powdered sinigang mix	1 sachet (44 g)	1,540	77%
Powdered adobo mix	1 sachet	1,200	60%
Pancit canton seasoning packet	1 packet	1,720	86%
INSTANT NOODLES AND READY MEALS			
Lucky Me Pancit Canton	1 pack (80 g)	1,720	86%
Nissin Cup Noodles	1 cup (60 g)	1,800	90%
Lucky Me Beef Mami	1 pack	1,520	76%
Nissin Yakisoba	1 pack	1,650	83%
CANNED GOODS			
Corned beef	½ cup (100 g)	950	48%
Spam	2 oz slice (56 g)	790	40%
Sardines in tomato sauce	½ can (70 g)	480	24%
Vienna sausage	4 pieces (60 g)	470	24%
Luncheon meat (CDO / Purefoods)	1 slice (30 g)	380	19%
Tuna flakes in oil, drained	½ can (70 g)	250	13%

Cured meats, ulam, fast food, snacks, and safer choices

Food / item	Serving	Sodium (mg)	% of 2,000 mg
CURED / PROCESSED MEATS AND FISH			
Daing na bangus	100 g	1,200	60%
Tinapa (smoked fish)	1 piece (80 g)	900	45%
Tapa	100 g	850	43%
Longganisa (sweet)	2 pieces (80 g)	750	38%
Tocino	100 g	670	34%
Hotdog (Tender Juicy / Purefoods)	2 pieces (90 g)	540	27%
Embutido	1 slice (60 g)	420	21%
COMMON ULAM AND CARINDERIA DISHES			
Pancit canton (carinderia plate)	1 plate (~300 g)	1,800	90%
Sinigang (with mix sachet)	1 bowl (~400 mL)	1,500	75%
Pancit palabok	1 plate	1,400	70%
Adobo (chicken / pork)	1 cup (~250 g)	1,200	60%
Kare-kare with bagoong	1 cup + 1 tbsp bagoong	1,800	90%
Lugaw with patis	1 bowl	800	40%
Tinola (no patis added)	1 bowl	520	26%
Inihaw na liempo (no marinade)	100 g	180	9%
FAST FOOD			
Jollibee Chickenjoy	1 piece	690	35%
Jollibee Spaghetti	1 plate	1,100	55%
McDo Crispy Chicken Fillet	1 piece	740	37%
McDo Big Mac	1 sandwich	970	49%
Mang Inasal (chicken thigh + rice)	1 meal	1,250	63%
KFC Original Recipe	1 piece	820	41%
SNACKS, BREAD, AND SAFER CHOICES			
Boy Bawang (garlic corn nuts)	1 pack (100 g)	600	30%
Pringles	1 small can (50 g)	530	27%
Sky Flakes crackers	4 crackers (30 g)	260	13%
Pandesal	2 pieces (50 g)	240	12%
Nova multigrain	1 pack (40 g)	180	9%
Plain steamed rice (kanin)	1 cup	< 5	< 1%
Fresh bangus, baked, no salt	100 g	75	4%
Boiled egg, no added salt	1 large	70	4%
Fresh fruit (any)	per serving	0-10	< 1%

The hardest truth in this table

One bowl of sinigang made with a sinigang mix sachet consumes three-quarters of your daily sodium budget. One pack of pancit canton nearly the entire day. The food itself is not the problem — it is the seasoning packets and condiment culture wrapped around it. Replace the packets, keep the dish.

Lower-sodium versions of the food you already cook

LOW-SODIUM COOKING — SMART SWAPS FOR THE FILIPINO KITCHEN

FLAVOR WITHOUT SALT	INGREDIENT SWAPS
Calamansi + garlic + onion + ginger = full flavor base.	Patis (1 tsp, 600mg) → Calamansi juice + herbs (2mg)
Use vinegar (suka) to add brightness without sodium.	Salt + MSG → Garlic powder + lemon zest
Toast and dry-fry spices to intensify flavor.	Canned tomatoes → Fresh tomatoes
Marinate in citrus and herbs instead of soy sauce.	Instant bouillon cube → Homemade broth (unsalted)
Cook rice without salt — add a pandan leaf instead.	Soy sauce → Reduced-sodium toyo (use half)
	Bagoong → Skip entirely — use anchovy-free sampaloc broth

Taste evolves — after 2–4 weeks, salty foods will taste too salty. Your palate adjusts.

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INSTEAD OF ...	TRY THIS ...
Patis on eggs / fried rice (1 tbsp = 1,420 mg)	→ Calamansi juice + crushed garlic + black pepper (~5 mg). Add a drop of patis at the end only if needed (~120 mg).
Sinigang mix sachet (1,540 mg)	→ Fresh sampalok pulp boiled and strained, or kamias, or unripe green mango. Add a small amount of patis at the end only (~150 mg total).
Knorr / Maggi cube in soup (1,400–1,500 mg)	→ Homemade broth: chicken bones + onion + ginger + leeks, simmered 1 hour, strained and frozen in portions. ~80 mg/cup.
Pancit canton with seasoning packet (1,720 mg)	→ Same noodles — throw out the packet. Sauté with garlic, ginger, cabbage, carrot, light low-sodium toyo (~400 mg total).
Corned beef breakfast (950 mg/serving)	→ Fresh ground beef sautéed with onion, garlic, tomato, calamansi, pepper (~120 mg).
Daing na bangus (1,200 mg)	→ Fresh bangus baked or grilled with calamansi, garlic, onion, pepper (~75 mg).
Adobo with full toyo (1,200 mg/serving)	→ Adobo with half the toyo + double the vinegar + extra garlic and bay leaf (~520 mg).
Hotdog, 2 pieces (540 mg)	→ Fresh chicken or pork strips marinated in calamansi / garlic / pepper, pan-grilled (~80 mg).
Toyomansi dipping sauce (820 mg/tbsp)	→ Pure calamansi + garlic + chili (~10 mg). Add 1 tsp toyo only if needed (~300 mg).
Bagoong on kare-kare (1,370 mg/tbsp)	→ Slow-roasted shrimp paste mixed with toasted garlic and chili — use ½ tsp instead of 1 tbsp (~230 mg).
Instant noodle breakfast (1,720 mg)	→ Lugaw made with homemade broth, ginger, and scallions; one boiled egg on top (~250 mg).

Reading nutrition labels — the 4 numbers that matter

● < 140 mg per serving — LOW

Buy freely. Two or three servings still keep you well under target. Look for this on canned goods, sauces, and snacks.

● 140–400 mg per serving — MODERATE

Acceptable in limited amounts (1–2 servings per day max). Account for it in the day's running total.

● 400–800 mg per serving — HIGH

One serving is 20–40% of your day's allowance. Use sparingly or find a lower-sodium alternative.

● > 800 mg per serving — VERY HIGH

Avoid for daily eating. A single serving consumes nearly half the day's budget. Reserve for rare occasions.

- **Servings per container** — a small can of corned beef lists "2.5 servings" — the sodium on the front is per serving, not per can. Multiply.

- **Reduced sodium ≠ low sodium** — it means 25% less than the original. If the original was 1,800 mg, "reduced" is still 1,350 mg.

How to make Filipino food taste Filipino — with less sodium

● Lean on acid

Calamansi, vinegar (suka), tamarind (sampalok), kamias, unripe green mango, sili leaves. Acid sharpens flavour the way salt does — without the sodium load.

● Use heat and aromatics

Sili labuyo, black pepper, ginger, lots of garlic, sibuyas, kutsay (leeks), tanglad, dahon ng laurel, oregano. Bloom them in oil first to release maximum aroma.

● Build your own broth

Save bones in the freezer. Simmer 2-3 hrs with onion, ginger, carrot, peppercorns. Strain, portion into 1-cup containers, freeze. Replaces every Maggi cube.

● Salt at the table, not the pot

Sodium on the surface tastes stronger than sodium dissolved through. A small pinch sprinkled at serving time equals a much larger amount added during cooking.

● Rinse canned goods

Drain canned tuna, sardines, beans, or vegetables into a strainer and rinse under running water for 60 seconds — removes 30-40% of the sodium.

● Grill, bake, inihaw

High-heat dry cooking concentrates natural flavour and reduces the need for added salt. Grilled bangus with garlic and pepper rivals any salted version for taste.

Hidden sodium — sources you would never suspect



Hidden source	Typical sodium	Why it is there
Pandesal / sliced bread	120-150 mg / piece	Salt controls yeast — 4-6 pieces a day adds up.
Breakfast cereals	180-250 mg / cup	Cornflakes, Special K — not necessarily low-sodium.
Processed cheese (Eden)	300-400 mg / slice	Sodium emulsifies the cheese and extends shelf life.
Pickles / atchara	500-700 mg / serving	Pickled in salt brine — even 'unsweetened' versions.
Frozen pre-marinated fish	400-800 mg / piece	Supermarket bangus soaked in salt water for shelf life.
Antacids (Maalox, Kremil, eno)	200-500 mg / dose	times daily. Sodium bicarbonate base — significant if taken several
3-in-1 coffee sachets	40-80 mg / sachet	coffee alone. Sodium emulsifiers in the creamer; 4 cups = 200+ mg from
Bottled salad dressing	200-500 mg / tbsp	suka, garlic. Preservative + flavour. Make your own with calamansi,
Fountain / isotonic drinks	40-200 mg / serving	label. Sodium is added for electrolyte balance. Always read the

Eating out and carinderia survival

● Order grilled and steamed

Inihaw na isda, tinolang manok (no patis), grilled liempo, pinakbet without bagoong, chopsuey, fresh lumpia, kanin. Typically 300-600 mg per serving.

● Avoid breaded, brined, and soup-based

Pancit (any), sinigang (broth holds all the sodium), kare-kare with bagoong, fast-food breakfast sets (corned beef / longganisa / tocino / hotdog). Often > 1,000 mg per plate.

● "Pakitabi lang po ang sauce."

Ask for toyo, bagoong, or gravy on the side and dip lightly. One meal can drop 600-800 mg of sodium by this single change alone.

● Skip or share the soup

Restaurant soup broths are the highest-sodium item on the menu. If you order soup, share it, eat the solids, and leave most of the broth.

Why "low-sodium salt" can be lethal in advanced CKD

❏ POTASSIUM CHLORIDE DANGER

Almost every commercial salt substitute (LoSalt, Pansalt, Morton Lite Salt, and most "heart-healthy" salts) replaces sodium with potassium chloride — 50-70% by weight.

One teaspoon contains 1,500–2,000 mg of potassium. In a CKD patient with impaired potassium excretion, this can produce life-threatening hyperkalemia within hours. Hyperkalemia at > 6.0 mEq/L is a cardiac arrhythmia emergency; the first symptom may be cardiac arrest.

AVOID COMPLETELY in CKD Stage 3b–5, in dialysis patients, and in anyone on an ACE inhibitor, ARB, or spironolactone.

Safe in early CKD Stage 1–2 ONLY if serum potassium is normal and monitored.

They are also hidden in many packaged "low-sodium" canned goods and sauces — always read the ingredient list. If "potassium chloride" appears, do not buy.

The safe alternatives

The safe flavour boosters are not salt substitutes at all — they are calamansi, vinegar, sampalok, garlic, ginger, sili, bay leaf, black pepper, kutsay, and tanglad: zero sodium, maximum Filipino flavour.

The 7-Day Reduction Plan

Going from a 5-gram day to a 2-gram day overnight sets most people up for failure. Replace the highest-impact sources first, in an order that builds momentum.

DAY 1

Move the salt shaker off the dining table to a cabinet. Add salt during cooking only. This single change typically cuts ~300 mg per day.

DAY 2

Replace breakfast patis with calamansi + garlic + pepper on eggs or fried rice. Reclaims ~1,400 mg every morning you would have used patis.

DAY 3

One day without instant noodles. Replace with rice + boiled egg + sautéed kangkong. Saves ~1,500 mg.

DAY 4

If you do eat instant noodles today, throw out the seasoning packet — sauté with garlic, ginger, and low-sodium toyo instead. Saves ~1,000 mg.

DAY 5

No bouillon cubes. Make a small batch of homemade chicken broth. Use it everywhere a recipe calls for water + Maggi/Knorr. Saves ~1,500 mg per cube avoided.

DAY 6

Read every label before buying at the grocery. Avoid anything > 400 mg per serving. Check the ingredient list for potassium chloride.

DAY 7

Eat out only once, choose well: grilled fish, inihaw, or a vegetable dish. Ask for sauce on the side. Skip the soup. Drink water, not soft drinks.

By the end of Week 1

Most patients who follow this plan reach 2,000–2,500 mg per day — already a meaningful improvement over the typical 5,000 mg starting point. Two more weeks of consistency and the new pattern becomes the default. Blood pressure responds within 3–4 weeks. Proteinuria responds within 6–8 weeks.

Sodium & Salt Reduction — Pocket Checklist

Kitchen substitutions to make this week

- Replace Maggi / Knorr cube with a small jar of dried onion + dried garlic + black pepper — use 1 teaspoon in soups and stews.
- Keep a bottle of reduced-sodium toyo within reach; move the regular toyo to the back of the cupboard.
- Buy 4–5 calamansi a week. Squeeze fresh into eggs, rice, soup, fish, even into water.
- Make and freeze one batch of homemade chicken broth. Use it everywhere a recipe calls for water + bouillon.
- Move the salt shaker off the dining table to discourage casual sprinkling.

Flavour boosters that add zero sodium

● Calamansi juice

Brightens any dish. Excellent on eggs, fish, rice, lugaw.

● Suka (vinegar)

The foundation of adobo and dipping sauces. Zero sodium.

● Garlic & ginger

Sauté in oil first to build deep base flavour.

● Sili (chili)

Any variety — labuyo, siling mahaba — adds heat with no sodium.

● Sampalok / kamias

Natural souring agents for sinigang — skip the mix.

● Black pepper

Ground fresh adds intensity. Pairs well with calamansi.

When to call your nephrologist — sodium red flags

- Blood pressure consistently > 140/90 mmHg despite medication — sodium load may be undermining the drugs.
- Sudden weight gain of > 2 kg in 1–2 days — sodium-driven fluid retention.
- Swelling (edema) of the feet, ankles, or face worsening week over week.
- Urine output noticeably decreasing — sodium restriction + diuretics need re-titration.
- Potassium level > 5.5 mEq/L — check for hidden potassium chloride in the diet.

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References: KDIGO 2024 CKD Guideline · WHO Sodium Intake Guideline 2012 · PURE Study, NEJM 2014 · He & MacGregor, Cochrane 2013 · Philippine FNRI Food Composition Tables.

MEDICAL DISCLAIMER — This handout is patient education material produced by W. G. M. Rivero, MD for general information only. It does not replace the advice of your own physician or dietitian. Sodium targets, dietary restrictions, and medication interactions vary by individual; always follow your nephrologist's specific instructions. Do not use potassium-based salt substitutes without explicit approval from your doctor.